

Quality Appraisal Criteria

The quality assessment framework employed five criteria, each scoring from 0 (No) to 1 (Yes), with 0.5 for “To some extent”:

1. **QA1:** Are the objectives and the context of research clear, and well connected with DKT?
2. **QA2:** Does the research adequately delineate the methodology (e.g., training dataset, the model architecture, and its hyper-parameters)?
3. **QA3:** Does the research adequately delineate the experiments (e.g., examining sequential or temporal consistency, assessing interpretability, addressing data inconsistencies, or evaluating the model’s real-world applicability)?
4. **QA4:** Is there congruence between methodology and methods used for data collection, analysis, and interpretation?
5. **QA5:** Are the study contributions and limitations clearly stated?

Studies scoring ≥ 3.5 points were classified as high quality, those between 1.5–3.5 points as medium quality, and those ≤ 1.5 points as low quality.

Distribution Summary

Total studies analyzed: 83

Quality distribution:

- High quality (≥ 3.5): 83 studies (98.8%)
- Medium quality (1.5–3.5): 1 study (1.2%)
- Low quality (≤ 1.5): 0 studies (0.0%)

Mean quality score: 4.32 (SD = 0.52)

Per-study Quality Appraisal Scores

ID	Study	QA1	QA2	QA3	QA4	QA5	Total	Quality
1	Calibrated Q-Matrix-Enhanced Deep Knowledge Tracing with Relational Attention Mechanism	1.0	1.0	1.0	1.0	1.0	5.0	High
2	Graph-based knowledge tracing: Modeling student proficiency using graph neural networks	1.0	1.0	1.0	1.0	1.0	5.0	High
3	Learning consistent representations with temporal and causal enhancement for knowledge tracing	1.0	1.0	1.0	1.0	1.0	5.0	High
4	Knowledge Graph and Personalized Answer Sequences for Programming Knowledge Tracing	1.0	1.0	1.0	1.0	1.0	5.0	High
5	Bridging the Vocabulary Gap: Using Side Information for Deep Knowledge Tracing	1.0	1.0	1.0	1.0	1.0	5.0	High
6	ELAKT: Enhancing Locality for Attentive Knowledge Tracing	1.0	1.0	1.0	1.0	1.0	5.0	High
7	Deep Knowledge Tracing with Learning Curves	1.0	1.0	1.0	1.0	1.0	5.0	High
8	Design and evaluation of Trustworthy Knowledge Tracing Model for Intelligent Tutoring System	1.0	1.0	1.0	1.0	1.0	5.0	High
9	Pull together: Option-weighting-enhanced mixture-of-experts knowledge tracing	1.0	1.0	1.0	1.0	1.0	5.0	High
10	Heterogeneous graph-based knowledge tracing with spatiotemporal evolution	1.0	1.0	1.0	1.0	1.0	5.0	High
11	Deep Knowledge Tracking Integrating Programming Exercise Difficulty and Forgetting Factors	1.0	1.0	1.0	1.0	1.0	5.0	High
12	Temporal learner modelling through integration of neural and symbolic architectures	1.0	1.0	1.0	1.0	1.0	5.0	High
13	A Knowledge Query Network Model Based on Rasch Model Embedding for Personalized Online Learning	1.0	1.0	1.0	1.0	1.0	5.0	High
14	Interpretable Knowledge Tracing with Multiscale State Representation	1.0	1.0	1.0	1.0	1.0	5.0	High
15	EKT: Exercise-Aware Knowledge Tracing for Student Performance Prediction	1.0	1.0	1.0	1.0	0.5	4.5	High
16	Knowledge tracing machines: Factorization machines for knowledge tracing	1.0	1.0	1.0	1.0	0.5	4.5	High

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ID	Study	QA1	QA2	QA3	QA4	QA5	Total	Quality
17	DF-EGM: Personalised Knowledge Tracing with Dynamic Forgetting and Enhanced Gain Mechanisms	1.0	1.0	1.0	1.0	0.5	4.5	High
18	Context-Aware Attentive Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
19	Tracking knowledge proficiency of students with calibrated Q-matrix	1.0	1.0	0.5	1.0	1.0	4.5	High
20	GameDKT: Deep knowledge tracing in educational games	1.0	1.0	0.5	1.0	1.0	4.5	High
21	Knowledge interaction enhanced sequential modeling for interpretable learner knowledge diagnosis in intelligent tutoring systems	1.0	1.0	0.5	1.0	1.0	4.5	High
22	Deep Graph Memory Networks for Forgetting-Robust Knowledge Tracing	1.0	1.0	0.5	1.0	1.0	4.5	High
23	Ensemble Knowledge Tracing: Modeling interactions in learning process	1.0	1.0	1.0	1.0	0.5	4.5	High
24	Federated Deep Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
25	Deep Learning vs. Bayesian Knowledge Tracing: Student Models for Interventions	1.0	1.0	0.5	1.0	1.0	4.5	High
26	A novel framework for deep knowledge tracing via gating-controlled forgetting and learning mechanisms	1.0	1.0	1.0	1.0	0.5	4.5	High
27	Infusing Expert Knowledge Into a Deep Neural Network Using Attention Mechanism for Personalized Learning Environments	1.0	1.0	1.0	1.0	0.5	4.5	High
28	Exploiting multiple question factors for knowledge tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
29	AdaptKT: A Domain Adaptable Method for Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
30	Dynamic Cognitive Diagnosis: An Educational Priors-Enhanced Deep Knowledge Tracing Perspective	1.0	1.0	0.5	1.0	1.0	4.5	High
31	Modeling knowledge proficiency using multi-hierarchical capsule graph neural network	1.0	1.0	1.0	1.0	0.5	4.5	High

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ID	Study	QA1	QA2	QA3	QA4	QA5	Total	Quality
32	Monitoring Student Progress for Learning Process-Consistent Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
33	SPAKT: A Self-Supervised Pre-TrAining Method for Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
34	Adaptive meta-knowledge dictionary learning for incremental knowledge tracing	1.0	1.0	0.5	1.0	1.0	4.5	High
35	GraphCA: Learning from Graph Counterfactual Augmentation for Knowledge Tracing	1.0	1.0	0.5	1.0	1.0	4.5	High
36	Enhanced Knowledge Tracing With Learnable Filter	1.0	1.0	1.0	1.0	0.5	4.5	High
37	Self-paced contrastive learning for knowledge tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
38	Global Feature-guided Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
39	MLC-DKT: A multi-layer context-aware deep knowledge tracing model	1.0	1.0	1.0	1.0	0.5	4.5	High
40	Interaction Sequence Temporal Convolutional Based Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
41	Integrating fine-grained attention into multi-task learning for knowledge tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
42	A probabilistic generative model for tracking multi-knowledge concept mastery probability	1.0	1.0	1.0	1.0	0.5	4.5	High
43	SFBKT: A Synthetically Forgetting Behavior Method for Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
44	Towards more accurate and interpretable model: Fusing multiple knowledge relations into deep knowledge tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
45	A Genetic Causal Explainer for Deep Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
46	Explore Bayesian analysis in Cognitive-aware Key-Value Memory Networks for knowledge tracing in online learning	1.0	1.0	1.0	1.0	0.5	4.5	High
47	Meta-path structured graph pre-training for improving knowledge tracing in intelligent tutoring	1.0	1.0	0.5	1.0	1.0	4.5	High

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ID	Study	QA1	QA2	QA3	QA4	QA5	Total	Quality
48	GELT: A graph embeddings based lite-transformer for knowledge tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
49	An efficient state-aware Coarse-Fine-Grained model for Knowledge Tracing	1.0	1.0	1.0	1.0	0.5	4.5	High
50	Heterogeneous Evolution Network Embedding with Temporal Extension for Intelligent Tutoring Systems	1.0	1.0	0.5	1.0	1.0	4.5	High
51	Leveraging Pedagogical Theories to Understand Student Learning Process with Graph-based Reasonable Knowledge Tracing	1.0	1.0	0.5	1.0	1.0	4.5	High
52	JKT: A joint graph convolutional network based Deep Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
53	Learning Process-consistent Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
54	Implicit Heterogeneous Features Embedding in Deep Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
55	Knowledge Query Network for Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
56	SGKT: Session graph-based knowledge tracing for student performance prediction	1.0	1.0	0.5	1.0	0.5	4.0	High
57	Knowledge structure enhanced graph representation learning model for attentive knowledge tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
58	Improving Knowledge Tracing with Collaborative Information	1.0	1.0	0.5	1.0	0.5	4.0	High
59	Enhancing Knowledge Tracing via Adversarial Training	1.0	1.0	0.5	1.0	0.5	4.0	High
60	Ability boosted knowledge tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
61	Assessing Student’s Dynamic Knowledge State by Exploring the Question Difficulty Effect	1.0	1.0	0.5	1.0	0.5	4.0	High
62	Enhancing Deep Knowledge Tracing with Auxiliary Tasks	1.0	1.0	0.5	1.0	0.5	4.0	High
63	Introducing Problem Schema with Hierarchical Exercise Graph for Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
64	What is wrong with deep knowledge tracing? Attention-based knowledge tracing	1.0	1.0	0.5	1.0	0.5	4.0	High

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ID	Study	QA1	QA2	QA3	QA4	QA5	Total	Quality
65	Learning Behavior-oriented Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
66	Knowledge tracing based on multi-feature fusion	1.0	1.0	0.5	1.0	0.5	4.0	High
67	Improving knowledge tracing via a heterogeneous information network enhanced by student interactions	1.0	1.0	0.5	1.0	0.5	4.0	High
68	DCKT: A Novel Dual-Centric Learning Model for Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
69	Transition-Aware Multi-Activity Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
70	Learning from Non-Assessed Resources: Deep Multi-Type Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
71	DKVMN-KAPS: Dynamic Key-Value Memory Networks Knowledge Tracing With Students' Knowledge-Absorption Ability and Problem-Solving Ability	1.0	1.0	0.5	1.0	0.5	4.0	High
72	Discovering Multi-Relational Integration for Knowledge Tracing with Retentive Networks	1.0	1.0	0.5	1.0	0.5	4.0	High
73	Improving Knowledge Tracing via Considering Students' Interaction Patterns	1.0	1.0	0.5	1.0	0.5	4.0	High
74	Stable Knowledge Tracing Using Causal Inference	1.0	1.0	0.5	1.0	0.5	4.0	High
75	Modeling Student Performance Using Feature Crosses Information for Knowledge Tracing	1.0	0.5	1.0	1.0	0.5	4.0	High
76	A deeper knowledge tracking model integrating cognitive theory and learning behavior	1.0	1.0	0.5	1.0	0.5	4.0	High
77	Attention and Learning Features-Enhanced Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
78	MonaCoBERT: Monotonic Attention Based ConvBERT for Knowledge Tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
79	Deep Knowledge Tracing Incorporating a Hypernetwork With Independent Student and Item Networks	1.0	1.0	0.5	1.0	0.5	4.0	High
80	Embedding cognitive framework with self-attention for interpretable knowledge tracing	1.0	1.0	0.5	1.0	0.5	4.0	High
81	Knowledge Tracing Model and Student Profile Based on Clustering-Neural-Network	1.0	1.0	0.5	1.0	0.5	4.0	High

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ID	Study	QA1	QA2	QA3	QA4	QA5	Total	Quality
82	Improving Interpretability of Deep Sequential Knowledge Tracing Models with Question-centric Cognitive Representations	1.0	1.0	0.5	1.0	0.5	4.0	High
83	Model-agnostic counterfactual reasoning for identifying and mitigating answer bias in knowledge tracing	1.0	1.0	0.5	1.0	0.5	4.0	High

Sensitivity Analysis

Given that the vast majority of studies (98.8%) achieved high quality scores and only one study was classified as medium quality, removing medium-quality studies would not substantially alter the key findings of this systematic review. The high mean quality score (4.32/5.0) indicates that methodological rigor in DKT research is generally strong across basic quality criteria.